RITIKA POOJARY

AI/ML Engineer | Full Stack Developer

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LinkedIn: ritika-poojary | Github - https://github.com/webritgithub | Portfolio

SUMMARY

Software Developer with hands-on experience in web development and AI/ML applications. Skilled in Python, JavaScript, SQL, HTML, and CSS, with experience in machine learning, computer vision, image/video processing, OCR, and API development. Experienced in building, testing, debugging, and deploying real-world software and AI solutions.

SKILLS

Programming Languages: Python, JavaScript | **Web Development:** HTML, CSS | **Database:** SQL

AI/ML: Machine Learning, Deep Learning, Computer Vision | **Generative AI:** LLMs, Prompt Engineering, Gemini API

Libraries & Tools: OpenCV, PyTorch, Pandas, NumPy, Roboflow, Visual Studio Code, Git, GitHub

EDUCATION

SIES Graduate School of Technology | B.E. - CSE | CGPA: 8.50 / 10 — **2025**
JVM New English School & Jr. College | 12th | MSBSHSE | Percentage: 86.67 / 100 — **2021**

WORK EXPERIENCE

AI/ML Intern | Aryahs World Infotech

May 2026 – October 2026

- Collect, prepare, and annotate datasets for computer vision and machine learning applications.
- Preprocess data and train, test, and evaluate machine learning models.
- Work on image/video processing, OCR, and computer vision-based solutions.
- Develop and integrate Python-based ML APIs with software applications.
- Assist with debugging, optimization, testing, and deployment of AI-based solutions.

Junior Software Developer | Advin Softwares Pvt. Ltd.

Oct, 2025 - Jan 2026

- Developed and enhanced CRM, ERP, and HRMS modules, including Sales Order, Job WorkOut (Billing), and Shift Master (HRMS), using Angular, Bootstrap, and CSS.
- Configured UIC framework to create dynamic forms, tables and modals while reducing repetitive code.
- Upgraded existing functionality and implemented client-specific customizations to meet business requirements.
- Designed and created backend tables and stored procedures to support module workflows and reporting.
- Collaborated effectively with the development team using GitHub for version control, merging, and feature integration, ensuring smooth deployment.
- Participated in requirement discussions, testing, and bug-fixing.

PROJECTS

Civic Issue Detection

May 2026 – June 2026

Python | PyTorch| ResNet18 | CBAM | OpenCV | Flask | Render

- Developed an 8-class civic issue image classification model using a pretrained ResNet18 with transfer learning, enhanced with CBAM attention to focus on relevant visual features.
- Created a custom dataset by manually collecting and organizing images from the internet, applying preprocessing, augmentation, and oversampling to improve class balance and model generalization.
- Improved real-world prediction reliability using a multi-zoom inference approach with majority voting and confidence-based tie handling, addressing incorrect predictions caused by small or off-center objects and complex backgrounds.
- Evaluated the model on real-world images, integrated it with a Flask web application, and deployed the application on Render for remote access.

Medical Emergency Detection System

June 2026 – July 2026

Python| PyTorch | X3D-S | OpenCV | Flask | HTML | CSS | JavaScript

- Developed a real-time fall detection system using the X3D-S video classification model, trained on the Le2i and UR Fall Detection datasets to classify video sequences as Fall or Normal.
- Integrated YOLOv8 Pose to detect body keypoints and analyze movement after a detected fall, using changes in keypoint positions to estimate whether the person remained still.
- Implemented a fall-to-emergency verification pipeline where a detected fall triggers a timer, and persistent stillness for the configured duration escalates the system status from Fall Detected to Emergency, helping reduce immediate false alerts.
- Tested the trained model with a live camera feed and integrated the detection pipeline into a real-time web interface displaying fall status, pose/stillness information, timer progression, and emergency status.

CERTIFICATIONS

- Web Development (CodSoft) — **20 Feb, 2024**
- Python Fundamentals for Beginners (Codchef) — **13 Jun, 2024**